Jaeseok Lee

Brown University

jaeseok_lee@brown.edu | Google Scholar | Personal Website

Research Interests

- Earthquake source physics; Fault-network geometry; Rupture dynamics; Statistical seismology; Ground-motion physics

Education

Ph.D. in **Earth, Environmental, and Planetary Sciences** (**Brown University**, *expected May 2027*)

- Advisor: Victor C. Tsai

M.S. in **Earth and Environmental Sciences** (**Seoul National University**, *Feb 2022*)

- Advisor: Junkee Rhie

B.S. in **Physics** and **Earth and Environmental Sciences** (**Seoul National University**, *Feb 2020*)

Publications

Manuscripts Under Review and in Preparation

- Tsai, V. C., **Lee, J.**, Aso, N., Hirth, G., & Faulkner, D. R. (*Under review*). Seismogenesis from the finite slip of elastically loaded geometric asperities.
- **Lee, J.**, Tsai, V. C., Hirth, G., & Trugman, D. T. (*In preparation*). Fault-network density as a geometric signature of earthquake high-slip patches.

Peer-Reviewed Research Articles

- **Lee, J.**, Tsai, V. C., Trugman, D. T., Hirth, G., & Chatterjee, A. (2025). Fault network geometry modulates earthquake source spectra across scales. *Geophysical Research Letters*, **52**(12), e2025GL115592. doi: 10.1029/2025GL115592
- **Lee, J.**, Tsai, V. C., Hirth, G., Chatterjee, A., & Trugman, D. T. (2024). Fault-network geometry influences earthquake frictional behaviour. *Nature*, **631**, 106–110. doi: 10.1038/s41586-024-07518-6
- Chatterjee, A., Trugman, D. T., Hirth, G., **Lee, J.**, & Tsai, V. C. (2024). High-frequency ground motions of earthquakes correlate with fault network complexity. *Geophysical Research Letters*, **51**, e2024GL109418. doi: 10.1029/2024GL109418
- **Lee, J.**, Rhie, J., & Cheon, H. S. (2024). Seismic hazard assessment in the southeastern Korean Peninsula for large earthquakes in northern Kyushu, Japan: A 3D numerical simulation of pseudodynamic rupture scenarios. *Bulletin of the Seismological Society of America*, **114**(5), 2734–2750. doi: 10.1785/0120230305
- **Lee, J.**, Song, J.-H., Kim, S., Rhie, J., & Song, S. G. (2022). Three-dimensional seismic-wave propagation simulations in the southern Korean Peninsula using pseudodynamic rupture models. *Bulletin of the Seismological Society of America*, **112**(2), 939–960. doi: 10.1785/0120210172

Comments and Replies

- **Lee, J.**, Tsai, V. C., Hirth, G., Chatterjee, A., & Trugman, D. T. (2025). Reply to: On the effects of fault alignment on slip stability. *Nature*, **642**(8068), E22–E23. doi: 10.1038/s41586-025-09118-4

Presentations

Invited Talks

- **Lee, J.**, Tsai, V. C., Hirth, G., Trugman, D. T., & Chatterjee, A. (Dec 2025). *A geometry-driven perspective on earthquake dynamics*. AGU Fall Meeting 2025, New Orleans, LA, United States.
- **Lee, J.**, Tsai, V. C., Hirth, G., Chatterjee, A., & Trugman, D. T. (Dec 2024). *The influence of fault-network geometry on earthquake frictional behavior*. AGU Fall Meeting 2024, Washington, DC, United States.
- **Lee, J.** (Jul 2024). *How fault network geometry affects earthquakes*. U.S. Geological Survey Earthquake Science Center Seminar, virtual.

Conference Presentations

- **Lee, J.**, Tsai, V. C., Hirth, G., & Trugman, D. T. (Apr 2026). *Investigating the geometric origin of slip asperities*. SSA Annual Meeting 2026, Pasadena, CA.
- **Lee, J.**, Tsai, V. C., & Peterson, M. D. (Apr 2025). *Improving earthquake ground motion models by incorporating fault network complexity* [Best Poster Award]. Brown DEEPS DIVE Research Symposium 2025, Brown University, Providence, RI.
- **Lee, J.**, Tsai, V. C., Hirth, G., Chatterjee, A., & Trugman, D. T. (May 2024). *Fault network geometry's control on earthquake rupture behavior*. SSA Annual Meeting 2024, Anchorage, AK.
- **Lee, J.**, Tsai, V. C., Hirth, G., Chatterjee, A., & Trugman, D. T. (Dec 2023). *The link between fault network geometry and fault creep behavior in California*. AGU Fall Meeting 2023, San Francisco, CA.
- **Lee, J.**, Tsai, V. C., Hirth, G., Chatterjee, A., & Trugman, D. T. (Dec 2023). *Understanding earthquake source spectral properties through fault network complexity*. AGU Fall Meeting 2023, San Francisco, CA.
- **Lee, J.**, Tsai, V. C., Hirth, G., Trugman, D. T., & Chatterjee, A. (Sep 2023). *Does fault network complexity explain stress drop variability?* SCEC Annual Meeting 2023, Palm Springs, CA.
- **Lee, J.**, Tsai, V. C., Hirth, G., Trugman, D. T., & Chatterjee, A. (Mar 2023). *Does fault network complexity explain stress drop variability globally?* Seismology Student Workshop 2023, Palisades, NY.
- **Lee, J.**, Song, J.-H., Kim, S., Rhie, J., & Song, S. G. (Dec 2021). *Three-dimensional seismic wave propagation simulations in the southern Korean Peninsula using pseudo-dynamic rupture models*. AGU Fall Meeting 2021, New Orleans, LA.
- **Lee, J.**, Song, J.-H., Kim, S., Rhie, J., & Song, S. G. (Aug 2021). *3D ground motion predictions of scenario earthquakes in the southeastern Korean Peninsula*. Asia Oceania Geosciences Society 18th Annual Meeting, online.
- **Lee, J.**, Song, J.-H., Rhie, J., & Song, S. G. (Dec 2019). *Prediction of ground motions in the southeastern Korean Peninsula for scenario earthquakes in northern Kyushu*. AGU Fall Meeting 2019, San Francisco, CA.

Workshops

- SCEC Fault Creep Workshop, San Jose, CA, Mar 2026.

Research Seminars

- DEEPS Geophysics Lunch Bunch Seminar, Brown University (Spring 2026). *How fault geometry shapes earthquake rupture and seismicity*.
- Seismology Research Group Seminar, University of Chicago (Spring 2025). *How fault-network geometry affects earthquakes*.
- DEEPS Geophysics Lunch Bunch Seminar, Brown University (Spring 2025). *Ground motion simulations for future scenario earthquakes*.
- Geophysics Research Group Seminar, Massachusetts Institute of Technology (Summer 2024). *Fault-network geometry influences earthquake frictional behaviour*.
- DEEPS Geophysics Lunch Bunch Seminar, Brown University (Spring 2024). *Fault-network geometry's control on earthquake rupture behavior*.
- DEEPS Geophysics Lunch Bunch Seminar, Brown University (Spring 2023). *Does fault-network complexity explain stress drop variability globally?*

Teaching Experience

Brown University, *EEPS 0050: Mars, Moon, and the Earth*, Fall 2026

Brown University, *EEPS 1690: Introduction to Methods in Data Analysis*, Fall 2024

Seoul National University, *3345.408: Seismology and Geodynamics*, Spring 2022

Seoul National University, *F36.103: Earth System Science*, Fall 2020

Fellowships and Awards

Brown University Dissertation Fellowship, 2026–2027

Best Poster Award, DEEPS DIVE Research Symposium, Brown University, 2025

Academic Excellence Scholarship, Seoul National University, 2020–2021

Brain Korea 21 Plus Research Scholarship, National Research Foundation of Korea, 2020

Presidential Science Scholarship, Korea Student Aid Foundation, 2013–2019

Professional Service and Leadership

Reviewer, *Geophysical Research Letters*; *Communications Earth & Environment*

Organizer, Brown Seismology Reading Group, 2024

Organizer, Brown Geophysics Lunch Bunch, 2023

Vice President, Brown Korean Graduate Student Association, 2025–2026

President, Brown Korean Graduate Student Association, 2024–2025